

Univerzitet u Beogradu
Elektrotehnički fakultet



Analytics for Data Products IDEs

-JETBRAINS INTERNSHIP PROJECT-

Student : Nikola Zlatanović 2024/3033

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Problem description:

In our IDE, tool windows provide quick access to specific tasks or information without leaving the editor. Users can open these tool windows either manually (using shortcuts, menus, or icons) or automatically (triggered by events like debugging or test failures).

Your task is to analyze the usage data for one particular tool window and investigate whether there's a noticeable difference in how long it stays open depending on whether it was opened manually or automatically.

Dataset description:

You'll be working with an event log that tracks tool window activity. Each row contains:

- user_id (anonymized user ID)
- timestamp (when the event occurred (in epoch milliseconds))
- event_id: "open" or "close"
- open_type: "manual" or "auto" (only present on open events; empty otherwise)

Like most real-world data, this isn't perfectly clean. You'll encounter some messy situations: you may see close events without a prior open, multiple opens in a row, some opens may not have a matching close before the dataset ends, etc.

Objective:

Your main objective is to figure out whether there's a meaningful difference in how long the tool window stays open depending on how it was opened. To do this, you'll need to:

- Match open/close pairs to reconstruct complete episodes of toolwindow usage
- Calculate how long the toolwindow was open during each episode
- Compare the duration patterns between manual and automatic opens
- Determine if any differences you find are statistically significant

Assumptions:

Several assumptions were made regarding the IDE logging behavior and user behavior:

1. Logger Behavior in the IDE:

- Opening a tool window is logged as "open", closing as "close".
- Hiding/unhiding a tool window is not logged.
- Multiple consecutive opens within a short time are all logged until the window is actually displayed; further open commands while the window is active have no effect and are not logged.
- Analogous behavior applies to closing.

2. IDE Settings:

- Default logging settings are assumed for all users (i.e., hiding/unhiding is not logged).

3. Unexpected Shutdowns:

- If the IDE is forcibly terminated (e.g., Task Manager, Alt+F4, shutting down the machine while IDE is running), no logs are recorded. Some users are assumed to occasionally use these methods.

4. User Errors:

- Users sometimes repeatedly open/close a window in a short time, which is logged multiple times.
- Occasional misclicks followed by quick closure are possible.

5. Per-user Pairing:

- Open/close events are matched per user, meaning logs from different users are treated independently.

6. Tool Window Type:

- The analysis focuses on the Debug tool window. This assumption is supported by observed patterns: short manual sessions vs. longer automatic sessions.

These assumptions were critical for pairing open/close events (one pair defines a session) and filtering irregular logs. Most assumptions were also informed by research on PyCharm IDE behavior and typical developer workflows (VM usage, keyboard vs. mouse preferences, etc.). Examples from the dataset are used to justify these assumptions.

Dataset overview:

- **Figure 1.1:** First few rows of the original dataset (timestamps converted to seconds).
- **Figure 1.2:** Dataset summary statistics: number of rows, number of open/close events, manual vs. automatic opens, number of users.
- **Figure 1.3:** Breakdown per user: total logs, open/close counts, manual vs. automatic opens.
- **Figure 1.4:** Preprocessed dataset: sorted by user_id and timestamp, timestamp offset relative to the first event in the group, additional columns hour and sec for human-readable time.

The dataset is relatively small (~3.5k rows). Open and close events are relatively balanced, while automatic opens are about twice as frequent as manual opens. Most users have 0–20 events; some prefer automatic opens, others manual. The processed dataset allows tracking the full sequence of tool window usage.

 ORIGINAL DATASET:

	timestamp	event	open_type	user_id
0	1752250204	opened	manual	1
1	1751836141	closed	<NA>	2
2	1752304475	closed	<NA>	3
3	1752498934	opened	auto	4
4	1752141991	closed	<NA>	5
5	1752308210	opened	auto	3
6	1752310292	opened	auto	3
7	1752276666	closed	<NA>	6
8	1752158089	opened	auto	5
9	1752174540	opened	auto	7
10	1751903461	opened	auto	8
11	1752185562	opened	manual	9
12	1752149417	closed	<NA>	10
13	1752147744	closed	<NA>	11
14	1752565444	closed	<NA>	12
15	1752250448	closed	<NA>	6
16	1752308784	closed	<NA>	3
17	1751599973	opened	auto	13
18	1751826936	opened	manual	2
19	1751836139	opened	manual	2

Figure 1.1

```
COUNTS OF EVENTS/USERS:
Number of logs: 3503
Number of opened logs: 1865
Number of closed logs: 1638
Number of manually opened logs: 661
Number of auto opened logs: 1204

Number of users: 205
```

Figure 1.2

NUMBER OF EVENTS PER USER:						
	user_id	logs_count	opened_count	closed_count	auto_count	manual_count
0	1	18	9	9	1	8
1	2	14	7	7	1	6
2	3	41	22	19	22	0
3	4	22	11	11	10	1
4	5	34	17	17	12	5
5	6	50	26	24	19	7
6	7	28	15	13	14	1
7	8	53	27	26	19	8
8	9	14	7	7	1	6
9	10	160	80	80	76	4
10	11	17	11	6	7	4
11	12	162	83	79	56	27
12	13	2	1	1	1	0
13	14	52	26	26	12	14
14	15	188	94	94	90	4
15	16	19	10	9	4	6
16	17	87	47	40	39	8
17	18	7	4	3	2	2
18	19	56	28	28	26	2
19	20	8	4	4	0	4

Figure 1.3

PREPROCESSED DATASET:						
	timestamp	event	open_type	user_id	hours	seconds
0	1751826102	opened	auto	1	0	0
1	1751826746	closed	<NA>	1	0	644
2	1751841258	opened	manual	1	4	756
3	1751849544	closed	<NA>	1	6	1842
4	1751985487	opened	manual	1	44	985
5	1751986623	closed	<NA>	1	44	2121
6	1752250204	opened	manual	1	117	2902
7	1752250204	closed	<NA>	1	117	2902
8	1752250205	opened	manual	1	117	2903
9	1752257432	closed	<NA>	1	119	2930
10	1752843408	opened	manual	1	282	2106
11	1752843421	closed	<NA>	1	282	2119
12	1752843424	opened	manual	1	282	2122
13	1752845073	closed	<NA>	1	283	171
14	1752877044	opened	manual	1	291	3342
15	1752884541	closed	<NA>	1	294	39
16	1753104440	opened	manual	1	355	338
17	1753106844	closed	<NA>	1	355	2742
18	1751656141	opened	auto	2	0	0
19	1751656199	closed	<NA>	2	0	58

Figure 1.4

Handling irregular logs:

Irregular logs are defined as follows:

- Consecutive open events for the same user without an intervening close
- Consecutive close events for the same user without a preceding open
- An open event at the very end of a user's logs or a close event at the beginning

Rationale

The rationale for this definition stems from both the behavior of the IDE logger and typical user behavior patterns:

1. Forced IDE Shutdowns / Crashes

- If a user forcibly terminates the IDE (e.g., Task Manager, Alt+F4, system shutdown) while the tool window is open, the close event is never logged.

- When the IDE is reopened, a new open is logged. This can lead to sequences like open ... open without an intervening close.
- 2. Repeated Window Launches**
- Users may click to open a tool window multiple times in rapid succession, either intentionally or by accident.
 - The logger records each open until the window is fully displayed. Sequences such as open open open close may appear.
 - If the user waits longer between clicks, these multiple opens can be separated by seconds or even minutes, but still lack corresponding closes, depending on their workflow.
- 3. Multiple Close Attempts / Misclicks**
- Users sometimes repeatedly click the close button (X) on a tool window.
 - This results in consecutive close events that are logged within seconds of each other.
- 4. Startup / Shutdown Artifacts**
- An open at the start or a close at the end of a user's logs can appear due to the IDE being terminated outside of normal workflow or due to the dataset not capturing the full session history.
 - These events cannot be paired reliably and would bias session duration calculations if included.
- 5. Noise from Virtual Machines or Remote Environments**
- Users working in VMs or remote desktops may have occasional logging inconsistencies due to system lag, slow I/O, or unsynchronized clocks.
 - This may produce small clusters of irregular opens or closes that do not represent true usage.
- 6. User Errors and Experimental Behavior**
- Some users experiment with opening and closing tool windows rapidly.
 - Some users prefer keyboard shortcuts while others prefer the mouse, leading to bursts of opens or closes that do not reflect standard usage patterns.

By defining irregular logs in this way and removing them, we ensure that session duration calculations reflect actual usage rather than artifacts of logging behavior, user errors, or unexpected IDE shutdowns.

Frequency and Characteristics

- Approximately **10% of all logs are irregular**, with the majority being consecutive open events.
- Consecutive close events are less frequent and usually occur within a few seconds, consistent with misclicks or repeated close attempts.
- Most users (110 out of total) have **no irregular logs**, while a smaller subset of users accounts for multiple irregular events.
- Irregular logs are spread unevenly across users, reflecting differences in workflow and IDE usage patterns.
- Visualizations in Figures 2.1–2.4 illustrate:
 - Examples of irregular sequences (open open close, open open open close, close close ...)
 - The distribution of irregular log counts per user
 - Characteristics such as time gaps between consecutive irregular events

These observations confirm that while irregular logs are present, they represent a minority of total events and can be systematically filtered out to ensure accurate session reconstruction and duration analysis.



IRREGULAR LOGS:

	timestamp	event	open_type	user_id	hours	seconds
0	1752317879	opened	auto	3	21	1703
1	1752512831	opened	auto	3	75	2255
2	1752517914	opened	auto	3	77	138
3	1752130885	opened	auto	4	46	2334
4	1752483323	closed	<NA>	4	144	1972
5	1752535753	opened	manual	6	177	2875
6	1752707767	opened	auto	6	225	2089
7	1752300741	opened	auto	7	35	622
8	1753264321	opened	auto	7	302	3002
9	1752492944	opened	auto	8	164	409
10	1751997243	opened	manual	11	18	975
11	1752196602	opened	manual	11	73	2334
12	1752196625	opened	auto	11	73	2357
13	1752496572	opened	auto	11	156	3504
14	1752956232	opened	auto	11	284	2364
15	1752212516	opened	manual	12	25	1927
16	1752212525	closed	<NA>	12	25	1936
17	1752213325	opened	auto	12	25	2736
18	1752213332	closed	<NA>	12	25	2743
19	1752460542	opened	auto	12	94	1553

Figure 2.1



COUNTS OF IRREGULAR LOGS:
 Number of irregular LOGS: 329, which is 9.39% of the total number of logs
 Total number of irregular 'opened' events: 278, which is 84.50% of all irregular logs
 Total number of irregular 'closed' events: 51, which is 15.50% of all irregular logs
 Total number of irregular 'auto opened' events: 212, which is 76.26% of all irregular opened logs
 Total number of irregular 'manual opened' events: 66, which is 23.74% of all irregular opened logs

Figure 2.2

COUNTS OF IRREGULAR LOGS BY USER:					
	user_id	total_irregular	irregular_open	irregular_close	
0	3	3	3	0	
1	4	2	1	1	
2	6	2	2	0	
3	7	2	2	0	
4	8	1	1	0	
5	11	5	5	0	
6	12	12	8	4	
7	14	2	1	1	
8	16	3	2	1	
9	17	7	7	0	
10	18	1	1	0	
11	22	3	2	1	
12	23	2	1	1	
13	24	9	9	0	
14	25	5	5	0	
15	26	3	3	0	
16	27	2	1	1	
17	28	12	7	5	
18	30	4	2	2	
19	31	1	1	0	

Figure 2.3

NUMBER OF USERS PER NUMBER OF IRREGULAR LOGS:						
DESCRIPTION: This table shows how many user have made 0/1/2/... irregular logs						
	num_irregular	num_users	total_opened	total_closed	manual	auto
0	0	110	0	0	0	0
1	1	27	27	0	12	15
2	2	26	33	19	18	15
3	3	19	50	7	13	37
4	4	3	9	3	3	6
5	5	6	28	2	9	19
6	6	4	22	2	1	21
7	7	1	7	0	0	7
8	8	3	16	8	1	15
9	9	1	9	0	0	9
10	10	1	10	0	0	10
11	12	2	15	9	6	9
12	17	1	17	0	0	17
13	36	1	35	1	3	32

Figure 2.4

Pairing logs:

Open/close events were paired per user, using only **consecutive open and close logs** in chronological order. For example, given the sequence:

o1 o2 o3 c4 c5

only o3–c4 is considered a valid session. The earlier o1 and o2 are treated as irregular because they likely correspond to sessions that were terminated unexpectedly (forced IDE shutdowns or system crashes) and do not have matching close events.

This pairing approach is consistent with the assumptions:

- Consecutive opens without a close indicate separate sessions with missing logs.
- A valid open/close pair (o3–c4) represents a complete session.
- **Figure 3.1:** Transformed dataset – each row corresponds to a single session, with session duration in seconds calculated as the difference between the close and open timestamps.
- **Figure 3.2:** Number of sessions per type (manual vs. automatic).
 - Automatic sessions are roughly **twice as frequent** as manual sessions, reflecting typical IDE behavior and debugging workflows.

TRANSFORMED LOGS:

	user_id	open_type	duration_sec	start_hours	start_seconds
0	1	auto	644	0	0
1	1	manual	8286	4	756
2	1	manual	1136	44	985
3	1	manual	0	117	2902
4	1	manual	7227	117	2903
5	1	manual	13	282	2106
6	1	manual	1649	282	2122
7	1	manual	7497	291	3342
8	1	manual	2404	355	338
9	2	auto	58	0	0
10	2	manual	313	0	210
11	2	manual	0	0	524
12	2	manual	1	45	2156
13	2	manual	3	47	1595
14	2	manual	2	49	3598
15	2	manual	1	327	2479
16	3	auto	1419	0	0
17	3	auto	4305	0	1906
18	3	auto	11074	1	2757
19	3	auto	34666	8	433
20	3	auto	17	17	2717
21	3	auto	23	17	2827
22	3	auto	38	18	1683
23	3	auto	13	18	2691
24	3	auto	23	18	2722
25	3	auto	453	18	2834

Figure 3.1

COUNTS OF SESSIONS:
 Total number of sessions: 1587
 Total number of manual sessions: 595
 Total number of auto sessions: 992

Figure 3.2

Analysis of durations of sessions:

Basic descriptive statistics were computed for both automatic and manual sessions, including **min, max, mean, standard deviation, median, percentiles, and number of outliers**.

- **Figure 4.1 & 4.2:** Summary statistics for automatic and manual sessions.
- **Figure 4.3:** Box plot comparing session durations.
- **Figure 4.4:** ECDF (Empirical Cumulative Distribution Function) for automatic vs. manual sessions.
- **Figure 4.5 & 4.6:** Histograms of session durations (linear and log scales).
- **Figure 4.7:** Mann–Whitney U test result comparing durations of automatic vs. manual sessions.

```
📊 AUTO durations (sec):
count      992.0
mean       6178.8
std        28314.0
min         0.0
25%         35.8
50%        185.0
75%       1196.2
max      409874.0

📊 MANUAL durations (sec):
count      595.0
mean       1505.3
std        10951.2
min         0.0
25%         2.0
50%        11.0
75%       135.5
max     180918.0
```

Figure 4.1

```
NUMBER OF OUTLIERS AND PERCENTAGE OF ALL SESSIONS PER TYPE:
Auto sessions: 154 outliers (15.52% of all auto sessions)
Manual sessions: 100 outliers (16.81% of all manual sessions)
```

Figure 4.2

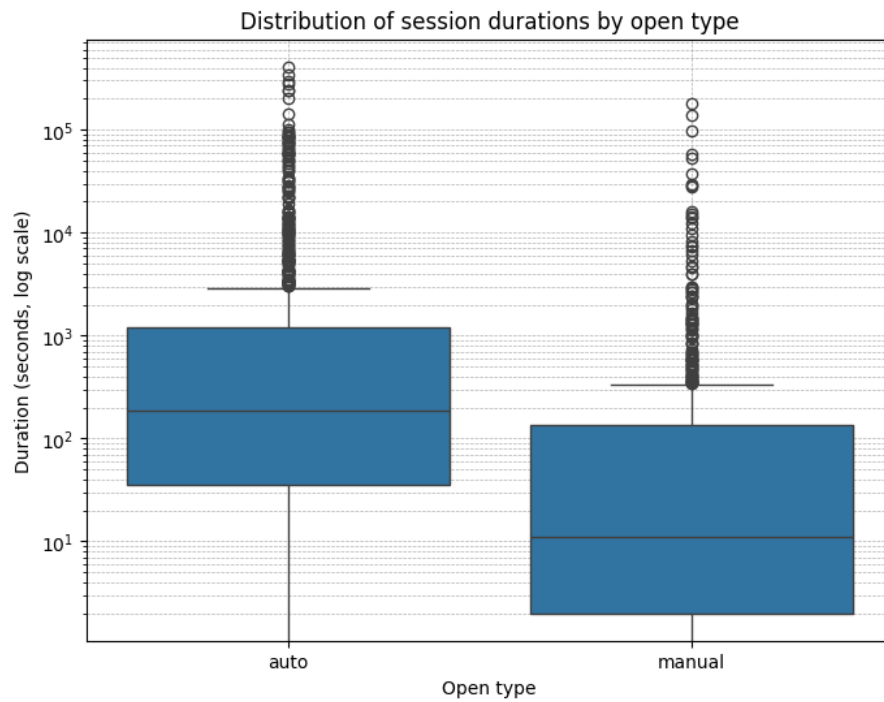


Figure 4.3

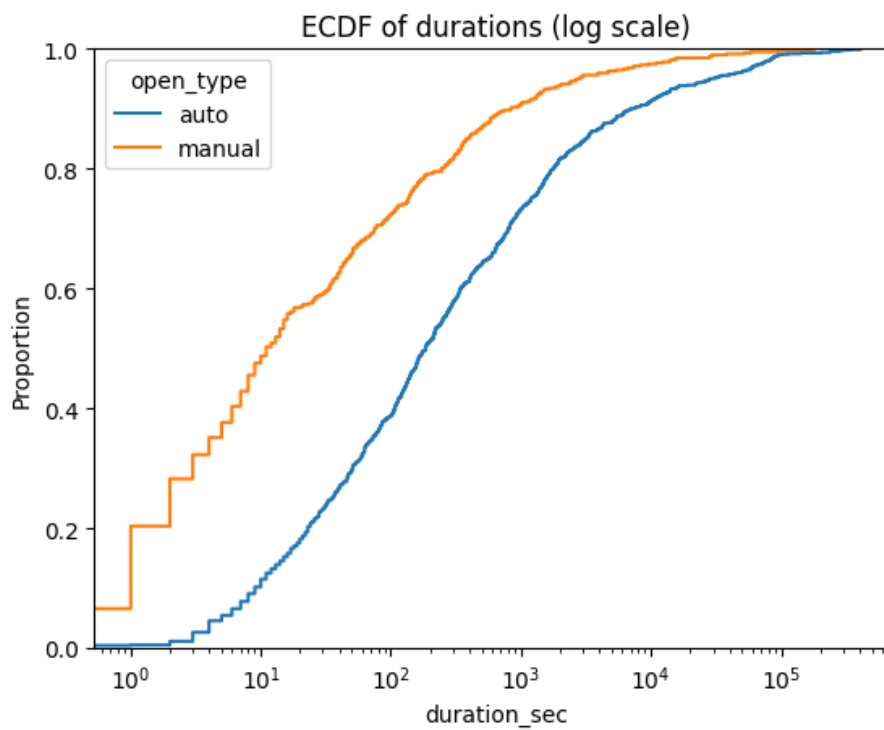


Figure 4.4

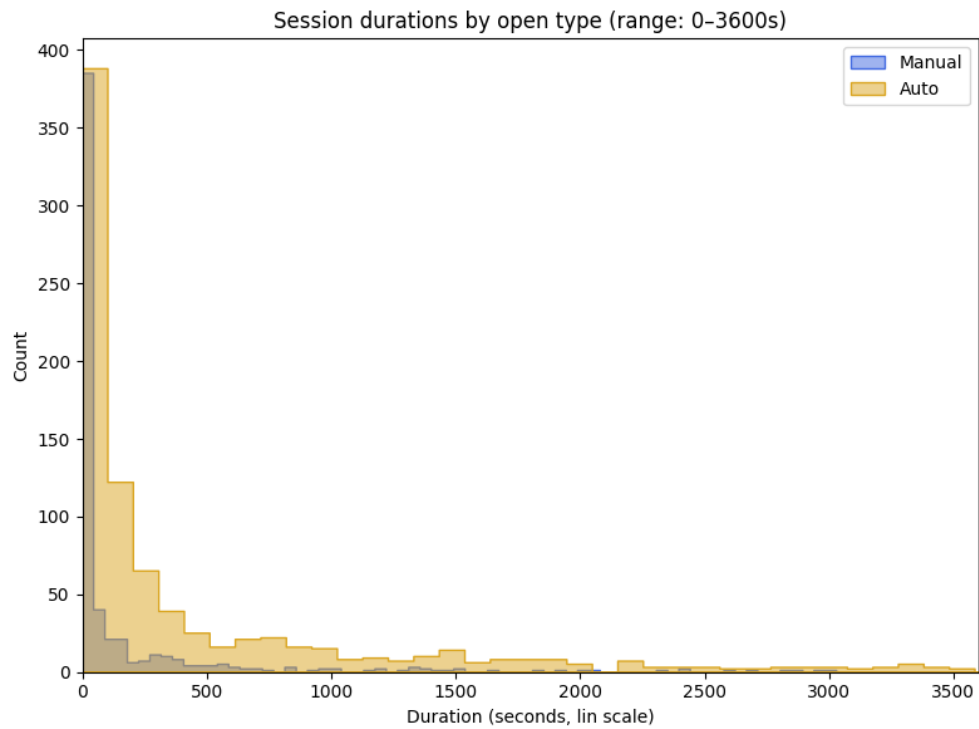


Figure 4.5

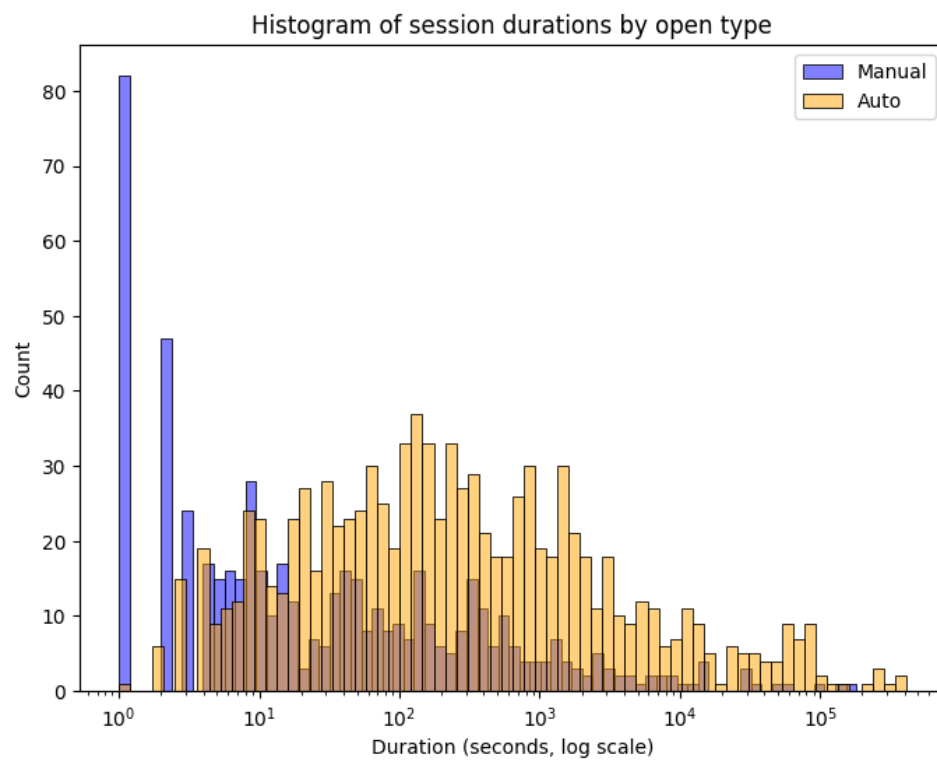


Figure 4.6

```
Mann-Whitney U test:  
U statistic: 442039.50, p-value: 4.5245e-62  
There is a statistically significant difference between auto and manual durations ( $p < 0.05$ ).
```

Figure 4.7

Observations:

From basic statistics, we observe a **clear difference** between automatic and manual sessions:

Metric	Automatic	Manual
Median	185 sec	11 sec
25th percentile	36 sec	2 sec
75th percentile	1200 sec	136 sec

- **Interpretation:** 75% of automatic sessions last longer than 36 seconds, whereas 50% of manual sessions are shorter than 11 seconds.
- **ECDF** shows a pronounced shift to the right for automatic sessions, indicating **longer durations**.
- **Histograms** suggest that both distributions are roughly exponential, but centered differently: automatic sessions have higher central values, while manual sessions are concentrated at very short durations.

The **Mann–Whitney U test** confirms this difference is statistically significant:

- **p-value $\approx 5e-62$** , effectively zero, indicating that the likelihood of these durations coming from the same distribution is virtually nonexistent.
- This result aligns with expectations:
 - Manual opens typically occur when a user wants to inspect something quickly, e.g., checking an error message.
 - Automatic opens often happen during debugging, which tends to last longer.
 - Occasional manual opens during debugging are rare relative to automatic opens, consistent with typical developer behavior.

Conclusion:

Based on the analysis of paired sessions and their durations, we can conclude the following:

- 1. Automatic sessions are significantly longer than manual sessions**
 - Median automatic duration is 185 seconds vs. 2 seconds for manual.
 - 75% of automatic sessions exceed 36 seconds, while 50% of manual sessions are shorter than 11 seconds.
- 2. Distributional characteristics differ**
 - Both distributions are roughly exponential, but the central tendency differs substantially.
 - ECDF and histograms indicate that automatic sessions are consistently longer across the entire range of durations.
- 3. Statistical significance is extremely high**
 - Mann–Whitney U test confirms the difference with $p \approx 5e-62$.
 - Non-normality of distributions is accounted for, and the test is robust for comparing medians.
- 4. Interpretation aligns with IDE and user behavior assumptions**
 - Manual opens reflect quick, user-initiated checks.
 - Automatic opens occur in the context of debugging, which requires longer attention spans.
 - Observed patterns validate the initial assumptions about IDE logging and user workflows.
- 5. Data quality considerations**
 - Irregular logs (~10% of total) were filtered, primarily consecutive opens, ensuring that only complete, meaningful sessions were analyzed.
 - Most users (110) had no irregular logs, suggesting that the dataset is generally reliable and the results are representative of typical behavior.

Overall, the analysis demonstrates a clear, statistically significant difference between manual and automatic tool window usage, providing insights into user behavior patterns within the IDE.